



Smart cooling fan controller, based on the LM56 temperature sensor IC, that turns the fan on at one temperature, then increases its speed if temperature rises above a second threshold

sources such as relays, radio transmitters, motors with arcing brushes, etc, as its wiring can act as an aerial and the internal junctions can act as rectifiers. In such cases, a small bypass capacitor from the power supply pin to ground rail helps clean up power supply noise.

Output filtering can be added as well. When using analogue sensors that should not directly drive large capacitive loads, the output filter capacitor can be isolated with a low-value resistor (like a zobel network) in series with the capacitor.

A three-terminal sensor needs three wires for power, ground and output signals. When sensing the temperature in a remote location, it is desirable to minimise the number of wires between the sensor and the main circuit board. In such situations you can use a two-terminal sensor. Moving to two wires means that power and signal must coexist on the same wires.

From audio amplifiers to personal computers

Audio amplifiers that dissipate more than a few watts always have their power transistors or the entire power amplifier IC bolted to a heat sink. It is often desirable to monitor temperature in an audio power amplifier to protect the electronics from overheating, either

by activating a cooling fan or shutting the system down. A good way to monitor the temperature is to mount the temperature sensor on the heat sink. Fit the sensor package by drilling a hole into the heat sink and cement the sensor to the heat sink with thermal-paste or heat-conducting epoxy.

Recent generations of personal computers dissipate a lot of power, which means they tend to get hot. High-performance computer processor chips consume excessive power and can get hot enough to suffer extremely harmful damage due to high temperature. To enhance system-stability it is often desirable to monitor processor's temperature and activate a cooling fan, slow down the system clock or shut down the computer completely if the processor gets too hot.

One good mounting site for the temperature sensor is in the centre of a hole drilled into the microprocessor's heat sink, which can be clipped to the processor or attached with epoxy. Another location is the cavity beneath a socketed processor. It is also possible to mount the sensor on the circuit board next to the microprocessor socket. •

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